ticks of edge.

Remark 11.16. So one could say either:

- "we got 25 cents of edge,"
- "we got 2 ticks of edge."

Both statements describe the same trade.

Example 11.13 (Corn Cash P&L)**.** To convert that edge into cash, note that 0.25 cents means

$$0.25 \times 0.01 = 0.0025 \text{ dollars.}$$

Then

$$0.0025 \times 5000 \times 20 = 250.$$

So the total theoretical cash P&L is

$$\boxed{\$250.}$$

Remark 11.17. A useful shortcut is to treat the 5000 bushel multiplier as 50 when working in cents instead of dollars. Then

$$0.25 \times 50 \times 20 = 250,$$

which gives the same answer more quickly.

12 Important Terminology and Electronic Marketplaces

We have already introduced many of the key market-making terms in earlier sections. This section revisits the most important ones from a more practical angle and then explains how electronic marketplaces organize speed, matching, and clearing.

12.1 A Basic Quoted Market

Example 12.1 (Reading a Simple Market)**.** Suppose a product is quoted as follows:

$$145 \text{ bid at } 930, \qquad \text{theo} = 945, \qquad 960 \text{ offered for } 240.$$

This means:

- 145 contracts are bid at 930,
- the desk's theoretical value is 945,
- 240 contracts are offered at 960.

Remark 12.1. The quoted market is the live trading environment. The theoretical value is not itself tradable, but it is the internal reference point against which a trader judges whether a bid or offer is attractive.

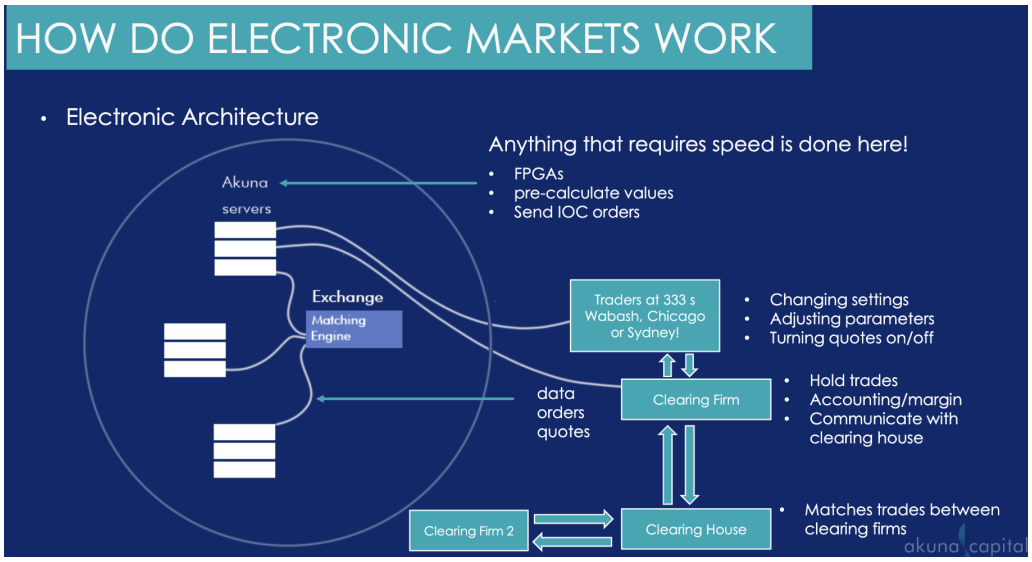

Figure 10: Electronic market architecture connecting trading systems, exchange matching engines, clearing firms, and the clearinghouse.

12.2 Review of Core Trading Terms

Remark 12.2. At the most basic level:

- the **bid** is where someone is willing to buy,
- the **offer** is where someone is willing to sell,
- the **size** is how many contracts are available,
- to **make a market** is to quote both a bid and an offer,
- the **spread** is the difference between the best bid and best offer.

Remark 12.3. Two other important broad terms are:

- **hedge**: a trade that reduces the risk of a larger position,
- **paper**: the outside customer flow trading against the market maker.

12.3 Market Participants Revisited

Remark 12.4. A **trader** is any active market participant. A **market maker** is a trader who continuously quotes two-sided markets and tries to earn the bid-ask spread while managing risk. A **local** is the older pit-trading term for a market maker, often historically an independent trader or small group concentrated in one pit.

Remark 12.5. A **broker** acts as an intermediary between buyers and sellers, especially on floors. Institutional participants such as hedge funds, banks, and asset managers often provide large outside orders, while retail customers tend to be smaller and less sophisticated.

12.4 Queue Priority in Practice

Definition 12.1 (Queue Priority)**.** **Queue priority** determines who trades first when multiple participants are bidding or offering at the same price.

Electronic markets commonly use **price-time priority**:

1. better price ranks first,
2. within the same price, earlier time ranks first.

Example 12.2 (Who Trades First?)**.** Suppose the order book contains the following bids:

$$\begin{array}{lcll} A & : & 12{:}43{:}01.00 & 1.25 \text{ bid } 30 \\ B & : & 12{:}43{:}01.02 & 1.25 \text{ bid } 20 \\ C & : & 12{:}43{:}01.05 & 1.25 \text{ bid } 250 \\ D & : & 12{:}25{:}00.00 & 1.24 \text{ bid } 500 \end{array}$$

Remark 12.6. The three orders at 1.25 are ahead of the order at 1.24 because price comes first. Among the three orders at 1.25, the execution order is A, then B, then C, because time comes next.

Example 12.3 (A Seller Arrives)**.** If someone now wants to sell 20 contracts at 1.25, they trade first with order A, since A is first in line at the best bid.

Remark 12.7. After that trade, order A is partially filled and still has 10 contracts remaining. Since it was only partially filled, it stays at the front of the queue for its remaining size.

12.5 Tick Size and Order Types Revisited

Definition 12.2 (Tick Increment)**.** The **tick increment** is the minimum allowed price change between one quoted price and the next.

Remark 12.8. If a market ticks in cents, then valid prices look like

$$100.01,\ 100.02,\ 100.03,\ \ldots$$

and prices in between are not valid.

Definition 12.3 (Immediate-or-Cancel)**.** An **immediate-or-cancel (IOC)** order is executed immediately for whatever amount can trade at the stated price or better. Any remaining quantity is canceled at once.

Remark 12.9. This makes IOC orders especially important in electronic trading, where speed matters and traders often want to interact with the current market without leaving residual exposure resting in the book.

Definition 12.4 (GFD and GTC)**.** A **good-for-day (GFD)** order remains active for the trading day unless executed or canceled. A **good-'til-canceled (GTC)** order remains active until filled or explicitly canceled.

Definition 12.5 (Liquidity)**. Liquidity** is the ability to buy or sell meaningful size without moving the market too much and without crossing an excessively wide spread.

Remark 12.10. Liquidity depends on both displayed size and market width. A product can be active, but still not be very liquid if size is thin or the spread is too wide.

12.6 How Electronic Markets Work

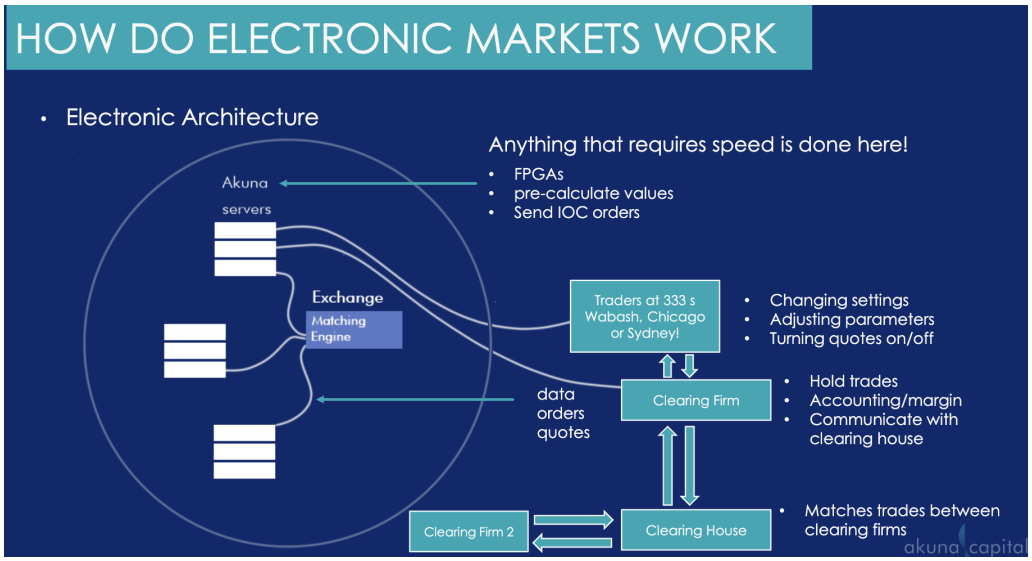

Figure 11: Electronic market architecture connecting trading systems, exchange matching engines, clearing firms, and the clearinghouse.

Remark 12.11. Electronic markets separate responsibilities across several layers:

- trading servers calculate values and send quotes or orders,
- the exchange matching engine matches orders,
- clearing firms record and support the firm's positions,
- the clearinghouse stands behind the matched trades across firms.

Remark 12.12. Anything requiring extremely low latency tends to happen as close to the exchange as possible. This includes rapid quote updates, pre-calculated values, and aggressive orders such as IOC orders.

Remark 12.13. Human traders still matter in electronic markets. They adjust settings, manage parameters, monitor risk, and decide when quotes should be turned on or off. The machines are fast, but the trading logic and risk controls are still supervised by people.

12.7 Why Standardized Contracts Matter

Remark 12.14. Before organized exchanges, producers and consumers often entered into private forward agreements. Those contracts had several weaknesses:

- **counterparty risk**: will the other side actually perform?
- **quality uncertainty**: what exactly will be delivered?
- **size mismatch**: can you find someone for your exact quantity?

Remark 12.15. Exchanges solve these problems through **standardization**. They specify the contract size, delivery rules, product quality, delivery locations, and trading increments. This allows one person to buy from any participant and another to sell to any participant without custom negotiation each time.

Remark 12.16. In practice, the exchange becomes the central trusted framework for the contract. This drastically reduces the risk and friction that would exist in a purely private market.

12.8 The Life of a Future

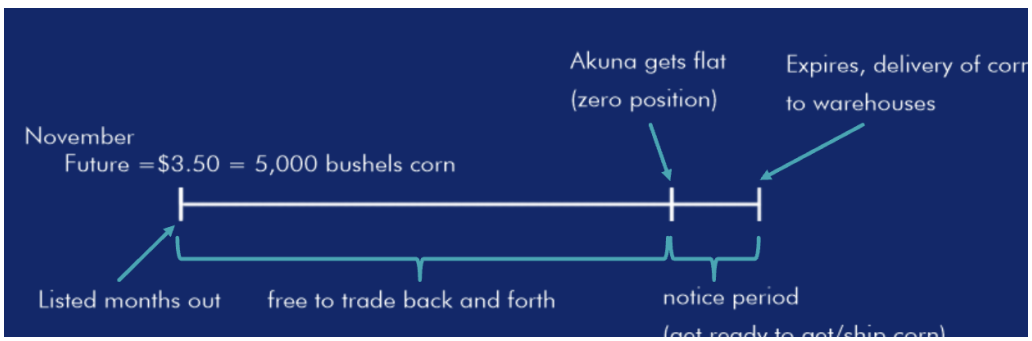

Figure 12: The life cycle of a futures contract: listing, trading period, notice period, and eventual settlement or delivery.

Definition 12.6 (Life Cycle of a Future)**.** A **futures contract** is listed before expiry, trades actively for a period of time, then enters a notice or delivery window, and finally settles either through cash settlement or physical delivery depending on the product.

Example 12.4 (Corn Future Timeline)**.** A November corn future may be listed months before expiration. During most of its life, participants are free to trade it back and forth. As expiry approaches, the contract enters a notice period. After that, participants still holding the contract may face physical delivery obligations.

Remark 12.17. Most market-making firms do not want physical delivery exposure. They flatten their positions before the delivery window becomes operational. In agricultural products such as corn, this means getting out well before any obligation to ship or receive actual grain.

Remark 12.18. This is one reason futures are so useful. Most participants can trade price exposure for months without any intention of ever delivering the commodity. The delivery mechanism is essential to the contract's credibility, but most traders exit before that stage.